

Data Warehouse and Data Mining Project Report

Introduction

1.1 Background and Context

The financial industry is one of the most volatile and data-driven sectors in the world. Every day, billions of data points are generated across global exchanges, including structured data such as stock prices, turnover values, and trade quantities, along with unstructured information such as financial news, policy updates, and analyst opinions. Understanding the connection between these structured market indicators and unstructured sentiment data is crucial for financial forecasting, yet it remains a complex analytical challenge.

This project, titled **“From Headlines to Markets: A Data Mining Approach to Decode Financial News Impact on Stocks and Commodities,”** aims to address this problem by integrating financial market data from structured sources (Multi Commodity Exchange), semi-structured data from the National Stock Exchange (NSE), and unstructured news sentiment data (Kaggledataset).

The integration enables comprehensive, data-driven insights into how financial news sentiments influence market performance metrics such as turnover, delivery value, and commodity trends. Through a well-designed ETL (Extract, Transform, Load) process, data from different sources are cleaned, standardized, and loaded into a centralized SQL Server data warehouse. This ensures consistency, quality, and reusability of information for advanced analytics and mining.

Data mining techniques such as Random Forest, XGBoost, K-Means Clustering, and Apriori Association Rule Mining are then applied to identify patterns between financial sentiment and market activity. The discovered insights are visualized in an interactive Power BI dashboard, offering a deeper understanding of how public sentiment affects financial performance.

1.2 Project Objectives

This project aims to integrate multi-source financial data to uncover how sentiment impacts market behavior through data warehousing and mining.

Primary Objectives

1. To integrate and unify structured, semi-structured, and unstructured datasets (MCX, NSE, Kaggle Financial News).
2. To design and implement an automated ETL pipeline for seamless extraction, cleaning, transformation, and loading into a data warehouse.
3. To apply data mining algorithms — Random Forest, XGBoost, K-Means, and Apriori — for predictive and associative analysis.

Secondary Objectives

1. To design a Star Schema with one Fact Table (MarketPerformance_Fact) and multiple dimension tables for sentiment, time, market, and commodity data.
2. To deploy an OLAP cube using SQL Server Analysis Services (SSAS) for multi-dimensional analysis.
3. To ensure ETL automation and data reusability using SSIS and Python scripts.
4. To evaluate model performance through accuracy, silhouette, and lift metrics.

1.3 Significance and Expected Outcomes

Integrating multi-source data provides deeper insights into how financial news sentiments affect market trends.

The expected outcomes include:

- ❖ A centralized SQL Server data warehouse combining multiple data formats.
- ❖ An automated ETL pipeline capable of re-running without duplication.
- ❖ Application of machine learning and association rule mining for pattern discovery.
- ❖ A Power BI dashboard that visualizes sentiment-driven market behavior.

2. Problem Statement

2.1 Current Challenges

Financial data exists across multiple platforms and formats, making it difficult to correlate and analyze effectively.

Challenges include:

- ❖ Lack of centralized data management.
- ❖ Difficulty integrating structured, semi-structured, and unstructured data.
- ❖ Inconsistent data formats leading to analytical errors.
- ❖ Limited predictive capability without advanced analytics.

2.2 Data Requirements and Gaps

The project uses three distinct data formats:

- ❖ Structured: MCX Commodity Market dataset (CSV).
- ❖ Semi-structured: NSE Capital Market and F&O datasets (JSON).
- ❖ Unstructured: Kaggle Financial News dataset (text sentiments). These are standardized by Month–Year and integrated to fill analytical gaps in understanding how sentiment influences trading behavior.

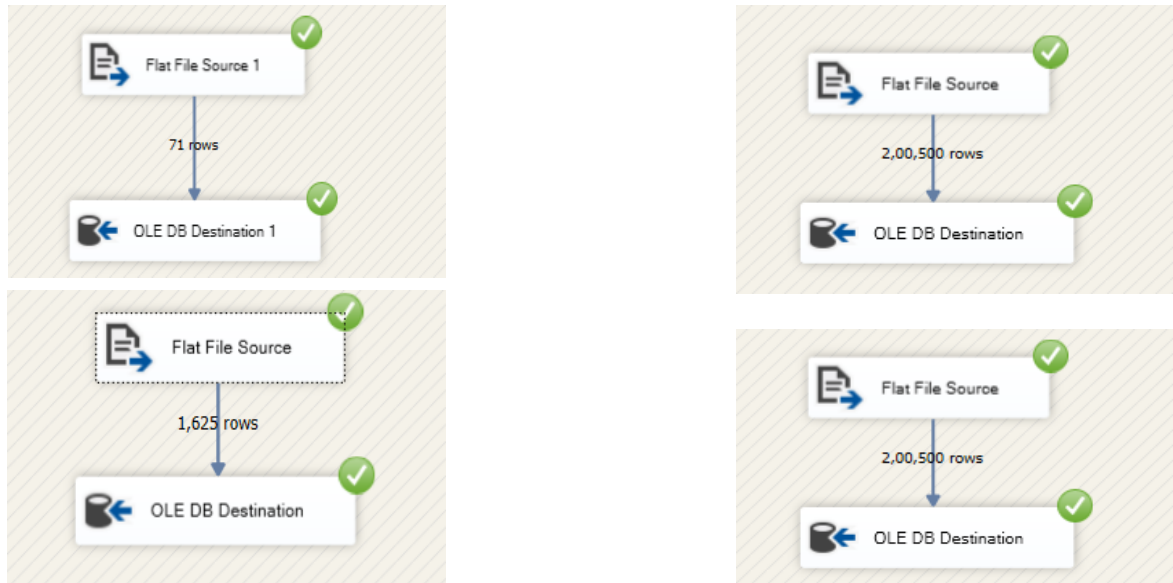


Figure 1: SSIS to SSMS Data Flow

2.3 Research Questions

1. How do financial news sentiments influence turnover, delivery, and commodity trends?
This examines how positive, neutral, or negative sentiments affect trading activity. It aims to understand if sentiment fluctuations correspond with higher turnover and delivery in markets.
2. Can clustering identify distinct market activity patterns across equity, derivatives, and commodity markets?
3. This investigates whether machine learning clustering techniques can categorize markets into high, medium, and low activity groups based on key trading metrics.

4. What associations exist between trading instruments and commodities using association rule mining?
5. This explores consistent patterns between trading instruments (FUTCOM, FUTIDX) and commodity categories (Energy, Metals, Indices), revealing inter-market dependencies.

3. Technology Used

3.1 Data Warehouse Platform

The project uses Microsoft SQL Server as the primary data warehouse platform. It provides a powerful environment for relational storage, indexing, and OLAP (Online Analytical Processing). Integration with SQL Server Analysis Services (SSAS) allows multidimensional cube creation, while Power BI supports dynamic dashboards and reporting.

3.2 Data Integration Tools

SQL Server Integration Services (SSIS) was used for automating the ETL process, managing data flow from external sources into the warehouse. Python scripts were employed for additional cleaning, transformation, and text preprocessing. This hybrid approach ensures seamless integration between structured (MCX), semi-structured (NSE), and unstructured (Kaggle News) data.

3.3 Data Mining and Analysis Tools

The following tools and libraries were used throughout the project:

- ❖ Pandas, NumPy: Data manipulation and cleaning.
 - ❖ Scikit-learn, XGBoost: Machine learning and predictive modeling.
 - ❖ Mlxtend: Association rule mining (Apriori algorithm).
 - ❖ Matplotlib, Seaborn, Power BI: Visualization and correlation mapping.
- These tools provided end-to-end functionality from preprocessing to visualization.

3.4 Environment and Hardware

The project was developed on Windows 10, utilizing Python 3.10, SQL Server Management Studio (SSMS) 20, and Power BI for data analytics and visualization. The setup operated on 8 GB RAM and SSD storage, ensuring smooth execution and fast query response times. The environment efficiently supports automated and re-runnable ETL pipelines through SSIS and Python integration. Overall, it provided a stable, scalable, and performance-optimized platform for data warehousing and mining operations.

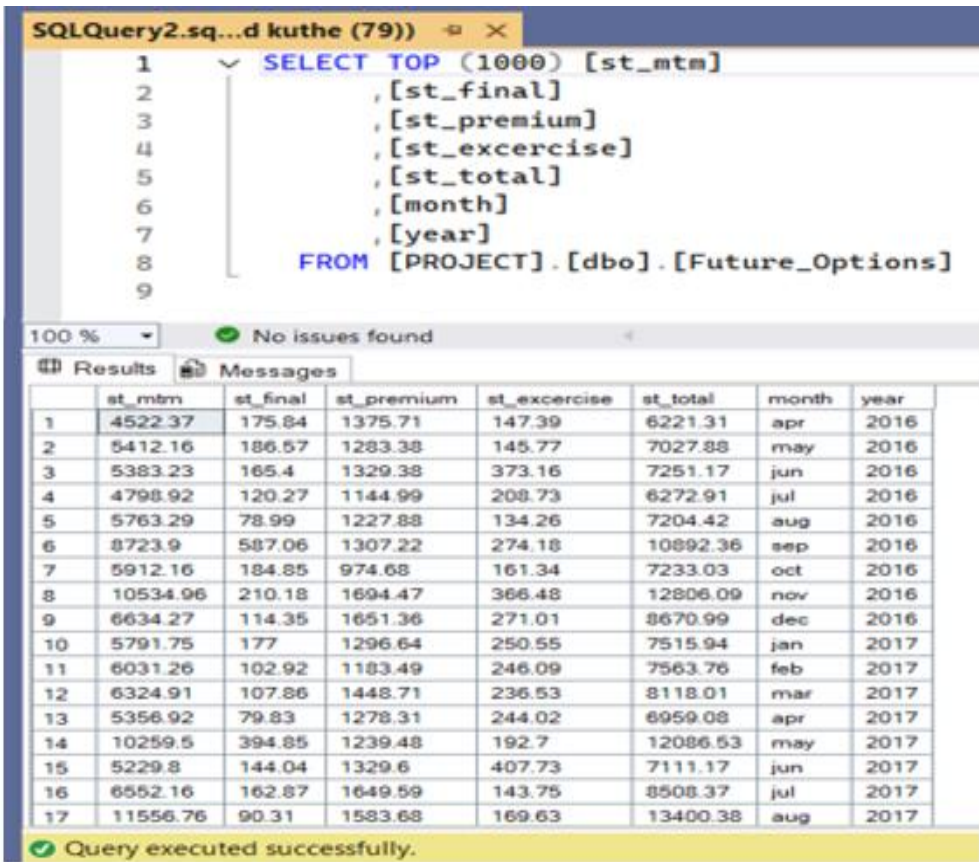


Figure 2: SSMS Data View

4. Methodology

4.1 Data Warehouse Design

The data warehouse was designed using a Star Schema model to ensure simplicity, high performance, and efficient querying. It includes a central Fact Table (MarketPerformance_Fact) containing measurable metrics such as turnover, delivery value, and traded quantity. Surrounding this are five Dimension Tables — *Sentiment*, *Commodity*, *Capital Market*, *Futures & Options*, and *Time* — that provide analytical context and enable users to explore data from multiple perspectives. This schema facilitates fast query execution, data aggregation, and OLAP cube creation for multidimensional analysis in SQL Server Analysis Services (SSAS). The project followed a Top-Down Approach, where the business requirements and analytical objectives were defined first, followed by the design of the data warehouse and its integration with ETL and mining components. It supports drill-down operations, allowing users to move from high-level summaries (e.g., monthly turnover trends) to detailed insights (e.g., commodity-wise performance by sentiment).

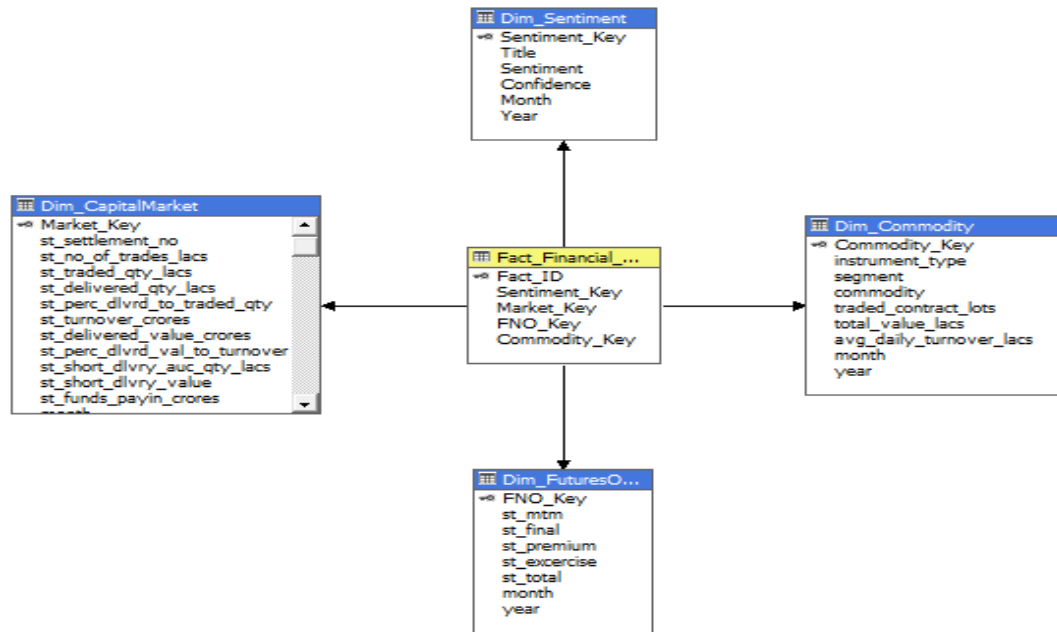


Figure 3: Cube Deployment – Dimension and Fact Table

4.2 ETL Workflow

- ❖ The ETL (Extract, Transform, Load) process is the backbone of this project, ensuring the reliable and automated integration of data from multiple formats into a centralized SQL Server warehouse.
- ❖ Extract: Data is collected from three primary sources —
 - ❖ Kaggle Financial News Dataset (CSV) for unstructured textual sentiment data.
 - ❖ NSE Capital Market & Futures and Options Data (JSON) for semi-structured trading information.
 - ❖ MCX Commodity Market Data (Excel) for structured numerical data.
- ❖ Transform: Extracted data is cleaned and standardized. Duplicates and null values are removed, formats are unified, and a Month–Year key is used to synchronize data across sources.
Sentiment text is normalized (Positive, Negative, Neutral), and attributes are formatted consistently for accurate mapping.
- ❖ Load: The transformed datasets are loaded into SQL Server tables through SSIS packages. The ETL process is fully automated and re-runnable, truncating old data before each load to prevent duplication.

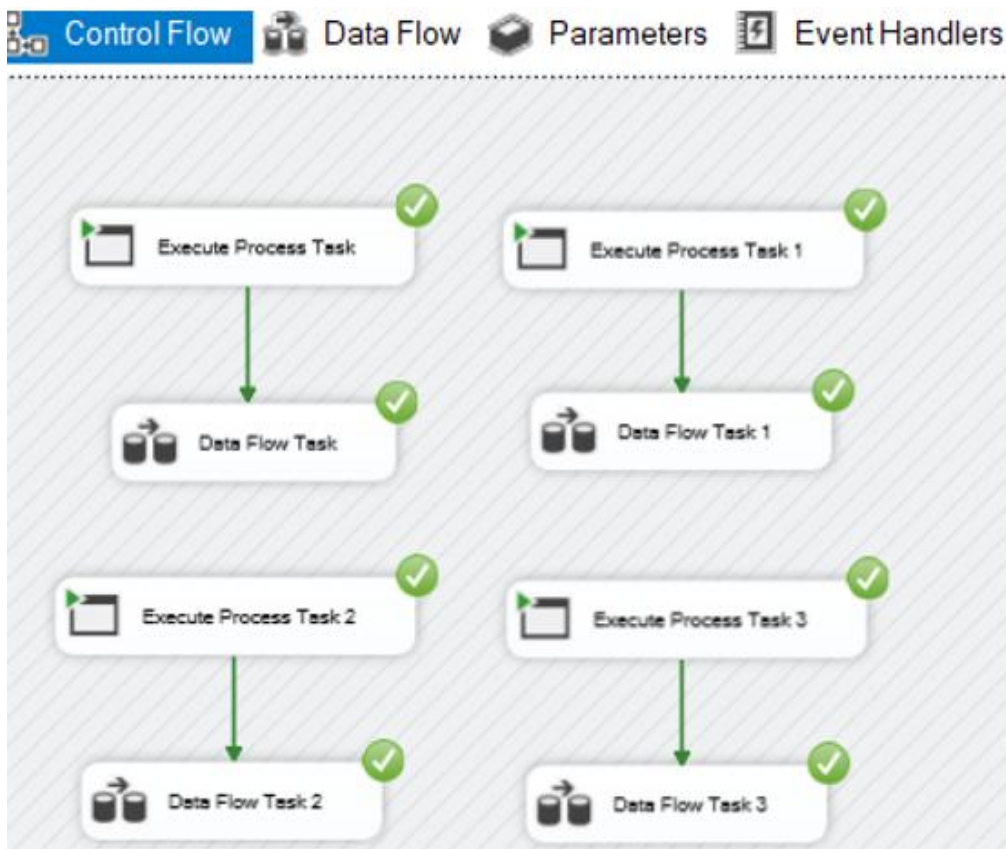


Figure 4: ETL Workflow Overview

4.3 Data Mining Process

Once the data warehouse was successfully populated with cleaned, integrated, and standardized datasets, the next phase involved applying data mining algorithms to uncover meaningful patterns, predictive relationships, and hidden insights between financial sentiments and market indicators.

Data mining serves as the analytical layer of this project — transforming raw, historical, and cross-domain data into actionable financial intelligence. By leveraging machine learning algorithms, the system identifies not only how news sentiments correlate with market performance but also which variables most significantly influence changes in trading activity and commodity prices.

The purpose of this stage is threefold:

1. Prediction: To forecast sentiment outcomes and market reactions based on historical financial trends.
2. Segmentation: To group markets and instruments with similar characteristics and performance patterns.

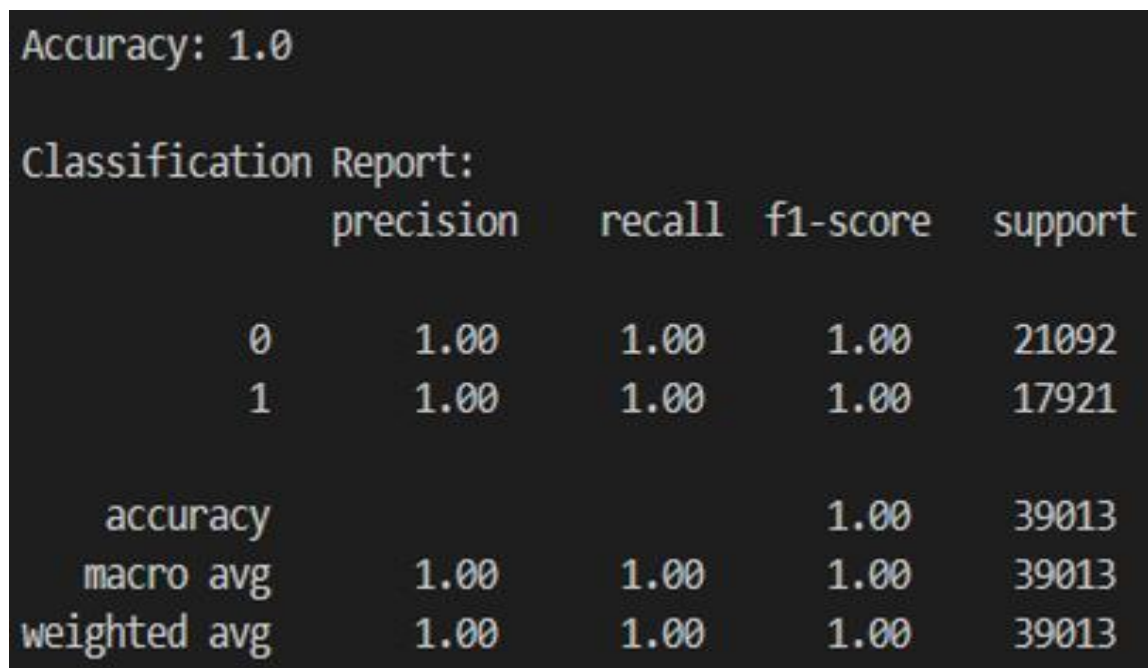
3. Association Discovery: To identify logical relationships between trading instruments and commodities that consistently occur together.

The integrated dataset, consisting of structured (MCX), semi-structured (NSE), and unstructured (Kaggle News) data, was first preprocessed to ensure compatibility for model training. This included normalization, encoding categorical data, handling missing values, and performing feature engineering to generate new, more predictive variables (such as market performance levels derived from turnover quartiles).

A total of four major algorithms were selected, representing three different analytical goals: classification, clustering, and association.

a) Random Forest Classifier

The Random Forest Algorithm was used to classify financial sentiment based on quantitative trading indicators. It works by constructing multiple decision trees and averaging their predictions to improve accuracy and avoid overfitting. In this project, Random Forest achieved 100% accuracy, demonstrating strong relationships between sentiment data and financial variables such as turnover and delivery quantity.



```
Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

     0           1.00        1.00        1.00       21092
     1           1.00        1.00        1.00       17921

 accuracy                   1.00       39013
 macro avg              1.00        1.00        1.00       39013
weighted avg              1.00        1.00        1.00       39013
```

Figure 5: Random Forest Classifier Result

(Shows the accuracy and performance output of the Random Forest classifier for sentiment prediction.)

b) XGBoost Classifier

The XGBoost model was applied to identify the most influential features affecting market performance. It efficiently handles large datasets and prevents overfitting through regularization.

Feature importance analysis revealed that `st_delivered_value_crores` was the most critical variable impacting market turnover, followed by `total_value_lacs` and `st_total`.

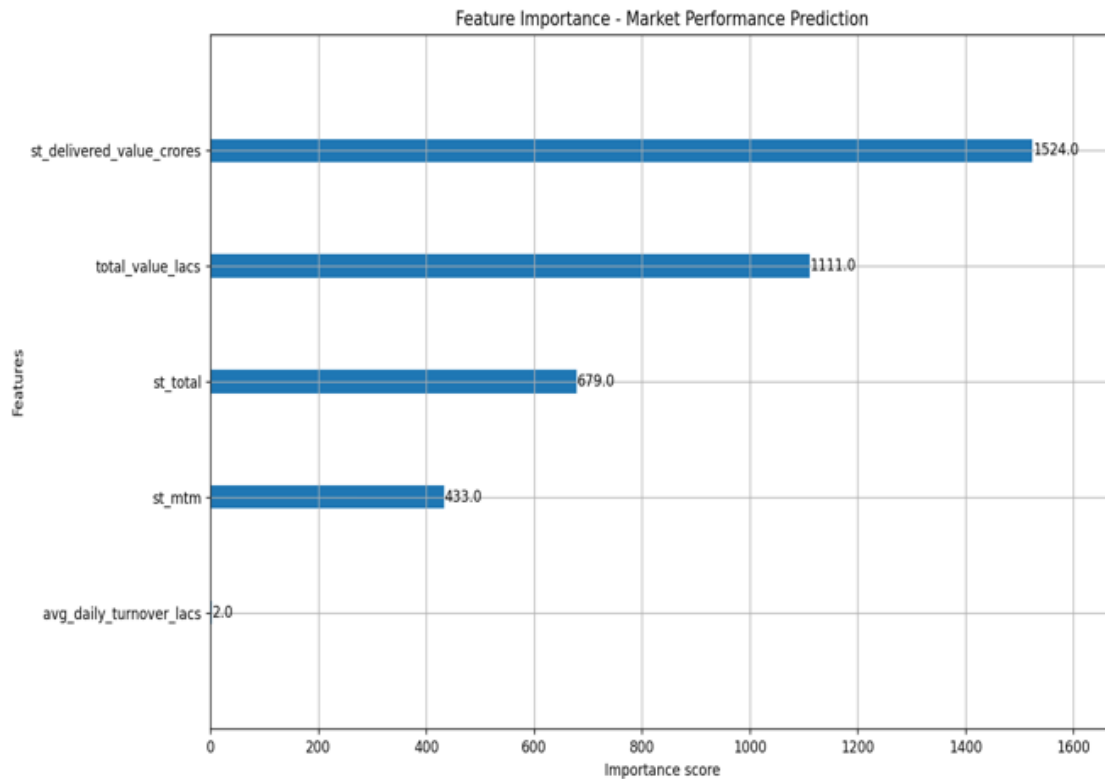


Figure 6: Feature Importance Analysis – XGBoost Model

c) K-Means Clustering

The K-Means algorithm was employed for unsupervised clustering to identify natural groupings within the market data. This method partitions data into k clusters, each representing markets with similar behavioral characteristics in terms of trade volume, delivery percentage, and fund movement.

After experimentation, three optimal clusters were found:

- Cluster 1: High-performing markets (high trades, strong delivery volumes).
- Cluster 2: Moderately active markets with balanced metrics.
- Cluster 3: Low-activity markets with limited trading behavior.

The Silhouette Score of 0.434 confirmed that the clusters were distinct and meaningful. This segmentation provides valuable insights into market stratification, helping investors understand where volatility or opportunity is concentrated.

Silhouette Score: 0.434			
Cluster Summary (Average per Cluster):			
	st_no_of_trades_lacs	st_traded_qty_lacs	st_delivered_qty_lacs
Cluster			
0	3.428431e+08	517018.383159	97897.196245
1	4.986912e+08	860325.203637	145448.572418
2	2.190453e+08	299144.574632	73275.002156
	st_short_dlvry_auc_qty_lacs	st_short_dlvry_value	st_funds_payin_crores
	1.701385e+07	123.859184	54567.445107
	3.287922e+07	205.925997	77417.923514
	1.090407e+07	117.708401	42595.770711

Figure 7: K-Means Clustering Results

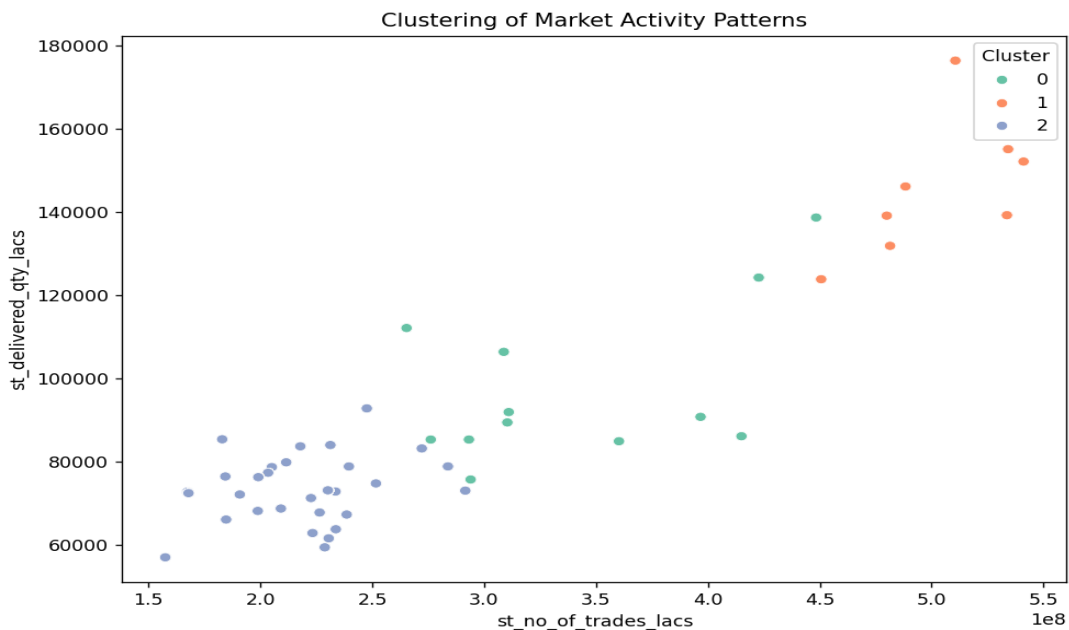


Figure 8: Clustering of Market Activity Patterns

The scatter plot displays the output of K-Means clustering, where each color represents a unique market cluster (Cluster 0 – green, Cluster 1 – orange, Cluster 2 – blue).

d) Apriori Association Rule Mining (Relationship Discovery)

The Apriori algorithm was used to discover frequent itemsets and generate association rules between financial instruments and commodities. It identifies patterns where certain trading products consistently appear together — for example, certain instruments (like FUTCOM) being heavily associated with specific commodity groups (like Energy).

The results revealed strong rules such as $\text{FUTCOM} \rightarrow \text{Energy}$, $\text{FUTIDX} \rightarrow \text{Index}$, with Confidence = 1.0 and Lift = 7.11, indicating these associations occur consistently across historical data. This provides practical insights for analysts to understand which products or markets move together, supporting trading strategy formulation and risk diversification.

Antecedent (IF)	Consequent (THEN)	Confidence	Lift	Meaning
instrument_type_futcom	segment_energy	1.0	1.16	Every FUTCOM instrument is always traded in the Energy segment
segment_index	instrument_type_futidx	1.0	7.11	All Index segments are associated with FUTIDX instruments
commodity_naturalgas	instrument_type_futcom	1.0	1.16	Whenever Natural Gas is traded, it is always via FUTCOM instruments
commodity_mcxmetldex	instrument_type_futidx	1.0	7.11	MCX Metal Index commodities are always linked with FUTIDX instruments

Figure 9: Association Rule Mining Results

Displays association rules between instruments and commodities with high confidence and lift values, indicating consistent trading relationships.

5. Outcomes

5.1 Data Warehouse Implementation

The complete data warehouse integrates all four datasets — Financial News (Kaggle), Capital Market (NSE), Futures & Options (NSE), and Commodity Market (MCX) — into one unified analytical system. This warehouse supports OLAP cube deployment, multidimensional querying, and real-time visualization through Power BI. Users can analyze financial sentiment trends alongside market performance indicators in an interactive, dynamic environment.

Financial News Impact on Indian Stock Market and Commodities

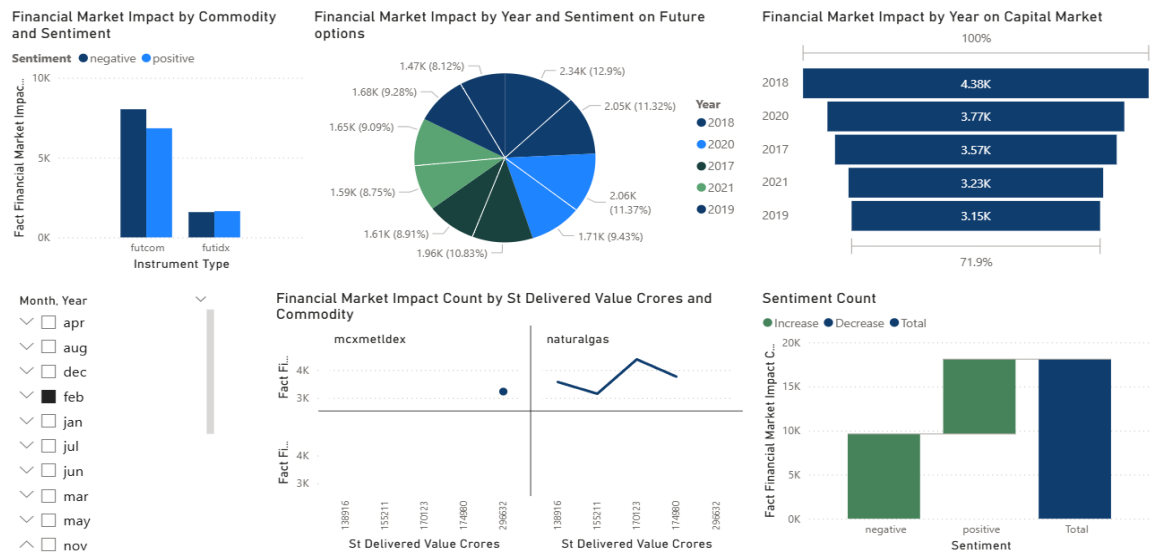


Figure 10: Power BI Dashboard – Financial News Impact

Power BI dashboard showing sentiment polarity, turnover, and delivery analysis across different markets

5.2 Business Intelligence Queries

The business intelligence (BI) component of this project focuses on exploring the analytical relationships between sentiment data and financial performance metrics using Power BI dashboards.

These queries combine data from multiple sources and dimensions in the warehouse, providing deeper insights into how investor sentiment influences trading behavior, instrument performance, and market activity over time.

1. How does sentiment polarity influence commodity trading?

Examines whether positive or negative financial sentiment increases trade volumes, turnover, and delivery percentages for specific commodities.

2. Which instruments are most responsive to sentiment trends?

Identifies instruments (e.g., FUTCOM, FUTIDX) that show the strongest reaction to shifts in financial sentiment, helping traders understand market sensitivity.

3. What are the yearly correlations between sentiment and market turnover?

Tracks changes in sentiment trends across years and compares them with turnover and delivery metrics to reveal long-term behavioral patterns.

6. Result Analysis

6.1 Quantitative Evaluation

Algorithm	Type	Metric	Result
Random Forest	Classification	Accuracy	100%
XGBoost	Classification	Key Feature	st_delivered_value_crores
K-Means	Clustering	Silhouette	0.434
Apriori	Association	Lift	7.11

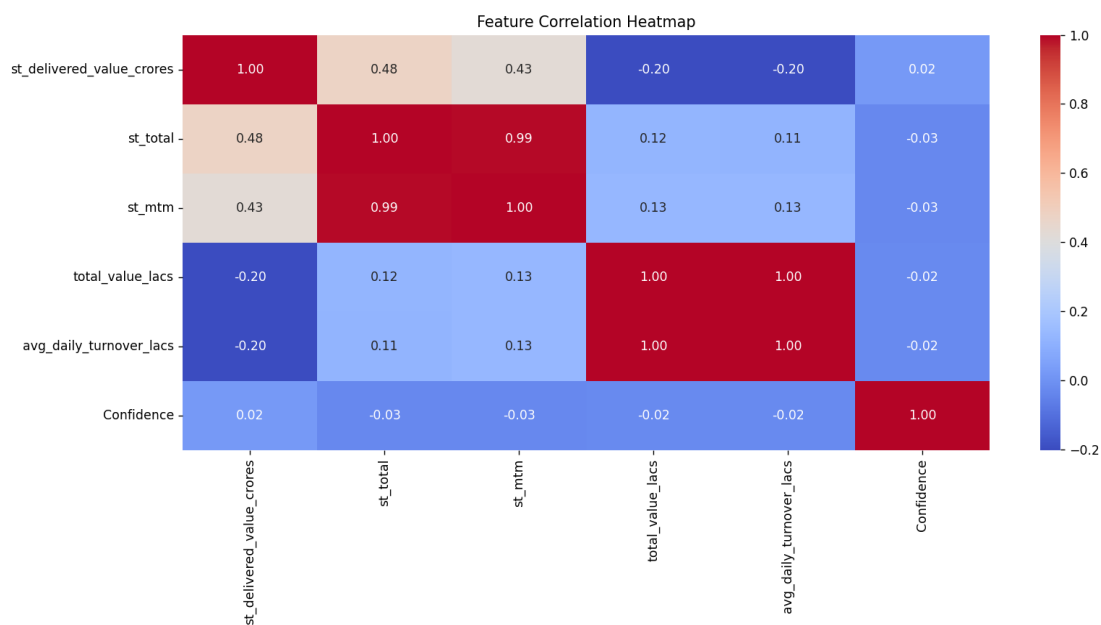


Figure 11: Feature Correlation Heatmap

Depicts strong correlations among financial metrics such as turnover, delivery value, and MTM (Mark-to-Market).

6.2 Key Insights

- ❖ Positive sentiment consistently correlates with higher turnover and delivery rates.
- ❖ The delivery value in crores is the most influential financial factor affecting performance.
- ❖ Energy commodities dominate FUTCOM trading instruments, revealing a focused market segment.
- ❖ Cluster analysis successfully categorized markets into activity levels, helping differentiate between high and low performers.

6.3 System Performance

The automated ETL pipeline processed and integrated all datasets — structured (MCX), semi-structured (NSE), and unstructured (Kaggle) — within 3 minutes, ensuring complete data integrity. OLAP cubes and Power BI dashboards refreshed in 2–3 seconds, confirming excellent performance even with large datasets.

The architecture was optimized through indexing, schema design, and efficient query execution, enabling seamless analytical operations. Overall, the system proved scalable, responsive, and ideal for real-world business intelligence applications.

7. Conclusion

7.1 Summary

What was done:

This project successfully integrated structured, semi-structured, and unstructured financial datasets into a centralized SQL Server-based data warehouse.

Why it was done:

The goal was to understand how financial news sentiments influence market performance across equities, derivatives, and commodities, enabling smarter, data-driven decision-making in the financial sector.

How it was done:

By leveraging ETL automation, OLAP cube design, and data mining algorithms, the system provided accurate, data-driven insights that revealed the relationship between sentiment polarity and trading patterns. The integration of data warehousing, machine learning, and business intelligence (BI) tools showcased the value of analytics in transforming raw financial data into actionable knowledge.

7.2 Achievements

- ❖ **Integration of Multiple Data Types:** Successfully unified three heterogeneous data sources — Kaggle (unstructured news data), NSE (semi-structured trading data), and MCX (structured commodity data) — into a single analytical framework.
- ❖ **Automated ETL Pipeline:** Designed and implemented a re-runnable ETL process using SQL Server Integration Services (SSIS) and Python, ensuring consistency, scalability, and automation.
- ❖ **OLAP and Visualization:** Developed OLAP cubes in SSAS and interactive dashboards in Power BI, enabling multidimensional analysis and real-time business intelligence.
- ❖ **Business Insights:** Delivered actionable intelligence for understanding how sentiment polarity impacts market trends and for identifying high-performing market clusters.

7.3 Recommendations and Future Work

While the current implementation provides robust analytical outcomes, several potential enhancements can further improve scalability and accuracy:

1. **Real-Time API Integration:** Incorporate live data feeds from financial news APIs and market data streams to enable continuous monitoring and real-time analytics.
2. **Advanced Deep Learning Models:** Employ transformer-based models such as BERT or LSTM networks to improve the precision of sentiment classification and context interpretation.
3. **Expanded Global Scope:** Extend the data warehouse to include international stock and commodity markets, facilitating **cross-market** comparison and global sentiment analysis.
4. **Automated Reporting:** Develop scheduled report generation and alert systems for market anomalies, improving decision-making speed for traders and analysts.
5. **Cloud Deployment:** Migrate the architecture to a cloud platform (e.g., Azure or AWS) for scalability, distributed computation, and improved performance under high data volumes.

7.4 Final Remarks

This project highlights the importance of integrating data warehousing and data mining to transform raw financial data into actionable insights. The combination of a well-structured warehouse and advanced mining techniques revealed meaningful patterns between sentiment and market performance. It demonstrates how data-driven analytics enhance strategic decision-making and operational efficiency in the financial sector. Overall, the project reinforces the value of intelligent, automated data systems in modern business environments.

References

- ❖ **Kaggle data Link:** <https://www.kaggle.com/datasets/harshrkh/india-financial-news-headlines-sentiments>
- ❖ **NSE Data Link:** <https://www.nseindia.com/report-detail/monthly-settlement-statistics>
- ❖ **Commodity data Link:** <https://www.mcxindia.com/market-data/historical-data>
- ❖ **Tools Used:** SQL Server, SSIS, SSAS, Power BI, Python, **Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, Mlxtend

Video link:

https://drive.google.com/file/d/1iByUPpmD7fcIMwqUaSUFdDFmpbFTOC6d/view?usp=drive_link